

Real Estate Valuation

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Statistical Learning

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Abstract

This report analyzes residential property values in Taipei, Taiwan, using the Real Estate Valuation dataset from the UCI Machine Learning Repository (<https://archive.ics.uci.edu/dataset/477/real+estate+valuation+data+set>) [cite: 1]. The study first explores the structure of the housing market through Principal Component Analysis (PCA) and clustering, revealing three economically meaningful groups of properties largely differentiated by accessibility to MRT stations, neighborhood amenities, and building age [cite: 1]. A supervised learning framework is then applied to assess the predictive power of structural and locational features on house prices [cite: 1]. Among the models considered—Linear Regression, Artificial Neural Networks (MLP), and Random Forest—the latter provides the best performance, confirming the importance of non-linear spatial effects [cite: 1]. Overall, the results highlight that proximity to MRT stations and geographical location are the strongest determinants of housing prices, while structural attributes such as house age play a secondary role [cite: 1]. These findings illustrate both the segmentation of Taipei’s housing market and the relevance of flexible models for price prediction [cite: 1].

1 Dataset Description and Preprocessing

The analysis is based on the Real Estate Valuation Data Set (Yeh, 2018), publicly available from the UCI Machine Learning Repository [cite: 1]. The dataset contains 414 observations of residential housing transactions in Taipei, Taiwan, collected between 2012 and 2013 [cite: 1]. Each record includes the transaction price per unit area (measured in New Taiwan Dollars per square meter) and several structural and locational characteristics [cite: 1].

The variables used in this project are [cite: 1]:

- **TransactionDate**: the year and month of the transaction, expressed as a decimal (e.g., 2013.250 $\approx$ March 2013) [cite: 1].
- **HouseAge**: the age of the property in years [cite: 1].
- **DistanceMRT**: the distance to the nearest mass rapid transit (MRT – metro) station, measured in meters [cite: 1].
- **ConvenienceStores**: the number of convenience stores in the living circle on foot [cite: 1].
- **Latitude** and **Longitude**: geographical coordinates of the property location [cite: 1].
- **PricePerUnitArea**: the target variable, defined as the unit price of the property (10,000 NTD/Ping, where Ping is a local unit, 1 Ping = 3.3 m²) [cite: 1].

Preprocessing

The dataset was subjected to preliminary cleaning before the analysis [cite: 1]:

1. **Missing values**: no missing values were found across the variables [cite: 1].
2. **Duplicated entries**: no duplicates were detected [cite: 1].

3. **Variable types:** all variables are numerical and therefore directly suitable for multivariate statistical analysis[cite: 1].
4. **Transaction Date:** this variable was not incorporated into the predictive models and was thus not subjected to additional transformation[cite: 1].
5. **Unit conversion:** the variable `PricePerUnitArea` was converted from 10,000 NTD/Ping into 10,000 NTD/m² to allow for standard interpretation and comparison[cite: 1].

Overall, the dataset is clean, well-structured, and ready for application of both unsupervised and supervised learning techniques[cite: 1].

2 Key Findings

- Three market segments emerge from PCA and clustering, mainly differentiated by proximity to MRT stations, housing age, and the availability of local amenities[cite: 1].
- Geographical location and distance to MRT stations are the strongest predictors of house prices, consistently dominating feature importance across models[cite: 1].
- Random Forest achieves the best predictive performance, outperforming the linear model and the neural network, suggesting the relevance of non-linear spatial relationships[cite: 1].

3 Methodology

The methodological approach of this project combines unsupervised and supervised statistical learning techniques[cite: 1]. The goal is to explore the structure of the housing market in Taipei and evaluate the predictive power of different models for housing prices[cite: 1].

3.1 Unsupervised Learning

Unsupervised learning methods are used to identify hidden structures in the dataset without relying on predefined labels[cite: 1]. Two techniques are applied[cite: 1]:

- **Principal Component Analysis (PCA):** used to reduce dimensionality and summarize the variance of the data into a smaller set of uncorrelated components[cite: 1]. This step helps to highlight the most influential variables driving heterogeneity among houses[cite: 1].
- **Clustering (*k*-means and hierarchical):** applied on the transformed feature space to detect homogeneous groups of properties[cite: 1]. Cluster solutions are evaluated using criteria such as the elbow method and silhouette scores[cite: 1].

3.2 Supervised Learning

Supervised learning methods are employed to predict the target variable (`PricePerUnitArea`) based on structural and locational features[cite: 1]. Several models are compared[cite: 1]:

- **Linear Regression:** baseline model, providing a simple interpretable benchmark[cite: 1].
- **Artificial Neural Network (ANN):** a simple feed-forward neural network (multilayer perceptron), included to test the performance of non-linear models inspired by machine learning[cite: 1].
- **Random Forest:** a tree-based ensemble method that captures nonlinear relationships and interactions among predictors[cite: 1].

3.3 Model Evaluation

For the supervised learning task, the dataset is split into training and test sets[cite: 1]. Models are evaluated using out-of-sample performance metrics[cite: 1]:

- Root Mean Squared Error (RMSE)[cite: 1]
- Mean Absolute Error (MAE)[cite: 1]
- Coefficient of Determination (R^2)[cite: 1]

These measures allow for a fair comparison of predictive accuracy across traditional statistical models and modern machine learning approaches[cite: 1].

4 Exploratory Data Analysis (EDA)

Before applying unsupervised and supervised learning methods, an exploratory data analysis was performed to examine the distribution of the target variable and its relationships with the main predictors[cite: 1].

4.1 Summary Statistics

Table 1 reports descriptive statistics for the variables used in the analysis[cite: 1].

The average unit price of housing is 115,000 NTD/m², with a median of 117,000 NTD/m², although values range widely (from 23,000 to 356,000 NTD/m²), indicating substantial price heterogeneity across properties[cite: 1]. Housing age displays a mean of 17.7 years, with a relatively wide range (0–43.8), suggesting the coexistence of both recently built and older structures[cite: 1].

Distance to the nearest MRT station also shows strong dispersion, with an average of 1,084 meters and a median of 492 meters, reflecting a right-skewed distribution and highlighting that many properties enjoy good accessibility, while a non-negligible group is located farther away (up to ≈ 6.5 km)[cite: 1]. The number of convenience stores within walking distance varies between 0 and 10, with an average of 4.1, pointing to meaningful differences in local amenities[cite: 1].

Table 1: Summary Statistics

Variable	Mean	Median	SD	Min	Max
PricePerUnitArea (10,000 NTD/m ²)	11.5	11.7	4.12	2.30	35.6
HouseAge	17.7	16.1	11.4	0.0	43.8
DistanceMRT (m)	1084	492	1262	23.4	6488
ConvenienceStores	4.09	4.0	2.95	0.0	10.0

4.2 Distribution of House Prices

The distribution of housing prices per square meter appears highly right-skewed, with most properties concentrated between 50,000 and 150,000 NTD/m², and a few extreme high-price outliers (Figure 1a)[cite: 1]. This asymmetry suggests that a log-transformation may be required for subsequent modeling[cite: 1].

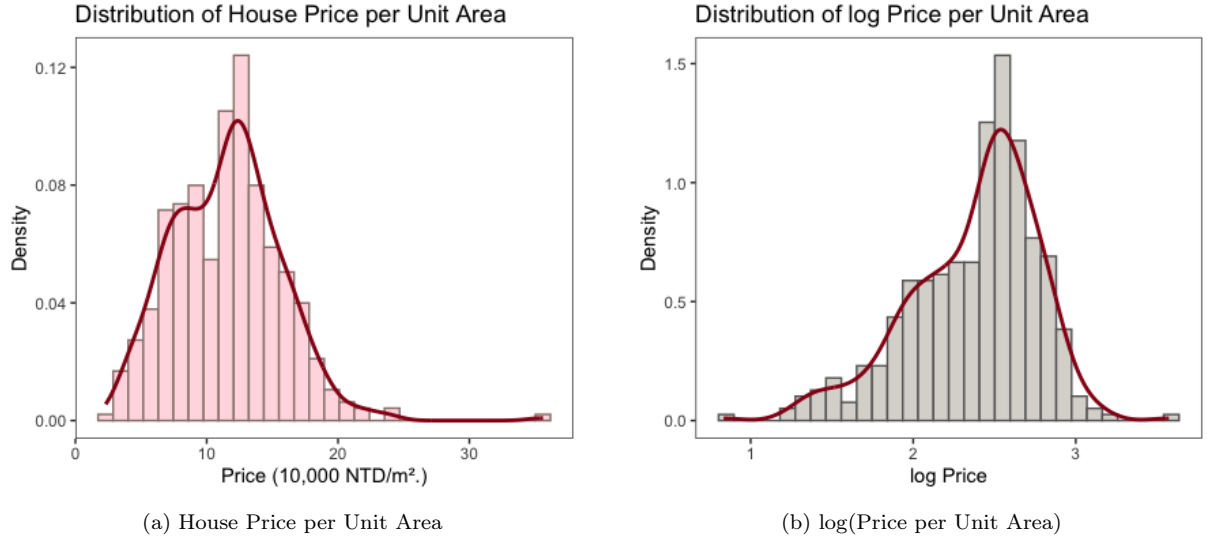


Figure 1: Distribution plots before and after logarithmic transformation.

After applying the logarithmic transformation, the price distribution becomes more symmetric (Figure 1b)[cite: 1]. This transformation reduces the impact of extreme outliers and stabilizes variance, making the data more suitable for regression and other supervised learning models[cite: 1].

4.3 Relationship with Location, Age, and Amenities

A clear negative relationship appears between housing prices and distance to the nearest MRT station (Figure 2a)[cite: 1]. Properties located closer to public transport hubs tend to have higher unit prices, highlighting the central role of accessibility in urban real estate valuation[cite: 1].

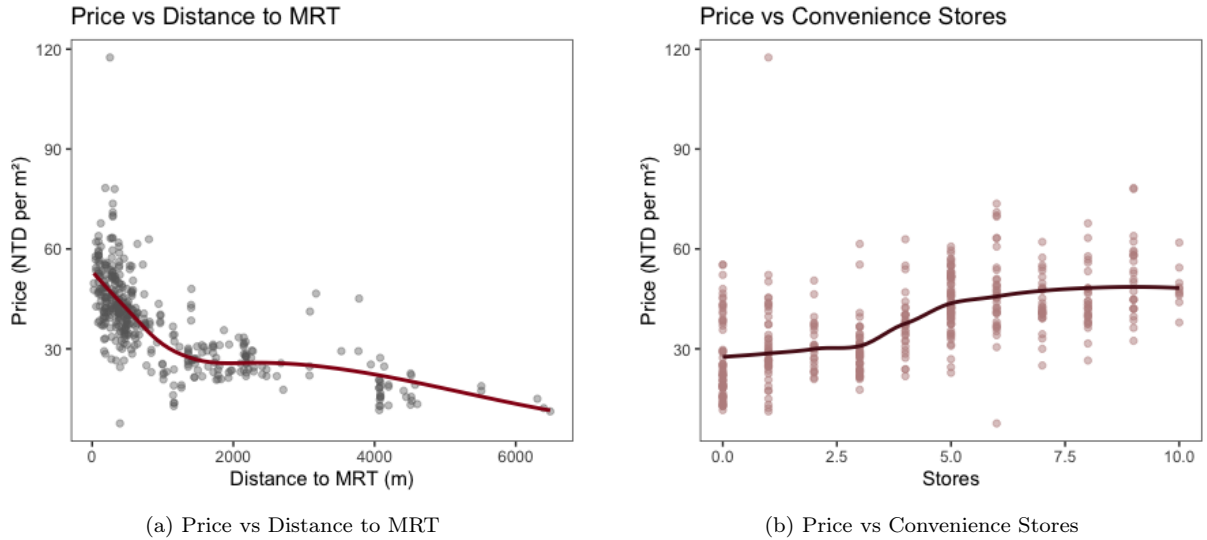


Figure 2: Bivariate relationships with accessibility and neighborhood amenities.

A positive association emerges between housing prices and the number of nearby convenience stores (Figure 2b)[cite: 1]. In addition, looking at building age (Figure 3), properties exhibit a non-linear U-shaped pattern where new buildings command higher prices, values dip for intermediate ages, and then slightly stabilize or rise for older properties in prime locations.

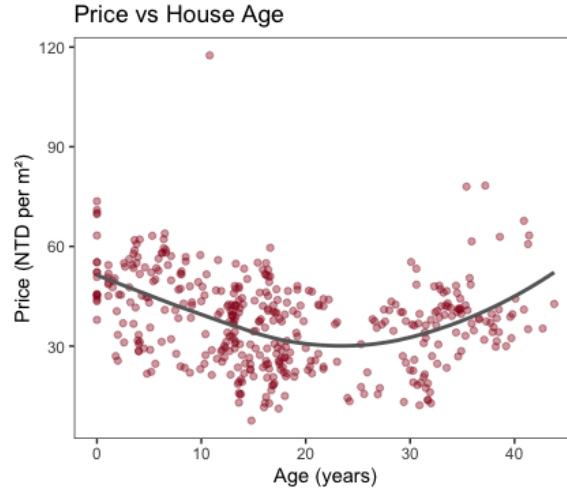


Figure 3: Price vs House Age.

4.4 Correlation Structure

The correlation matrix (Figure 4) confirms the main bivariate patterns: unit price is negatively correlated with distance to MRT (-0.67) and positively correlated with both convenience stores ($+0.57$) and geographical coordinates (latitude $+0.55$, longitude $+0.52$) [cite: 1]. These results support the idea that location and neighborhood services are key determinants of housing prices in the dataset [cite: 1].

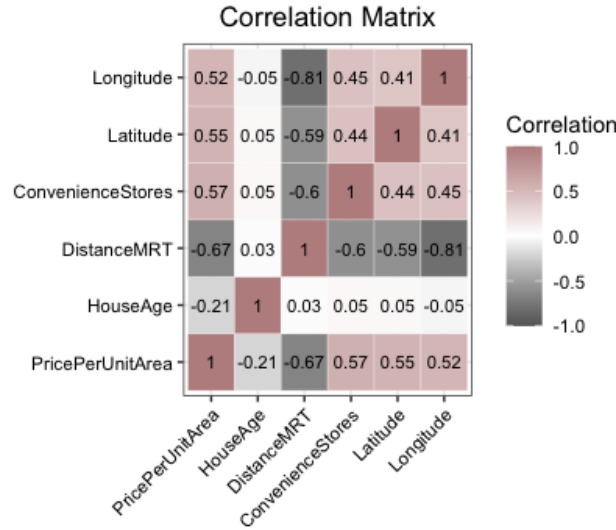


Figure 4: Correlation Matrix of Numerical Variables.

5 PCA Results

The Principal Component Analysis (PCA) was applied to five predictors (HouseAge, DistanceMRT, ConvenienceStores, Latitude, and Longitude) [cite: 1]. The eigenvalue decomposition shows that the first two principal components explain 74% of the total variance (PC1 : 53.5%, PC2 : 20.4%) [cite: 1]. Given this high cumulative proportion, the analysis focuses on PC1 and PC2 [cite: 1].

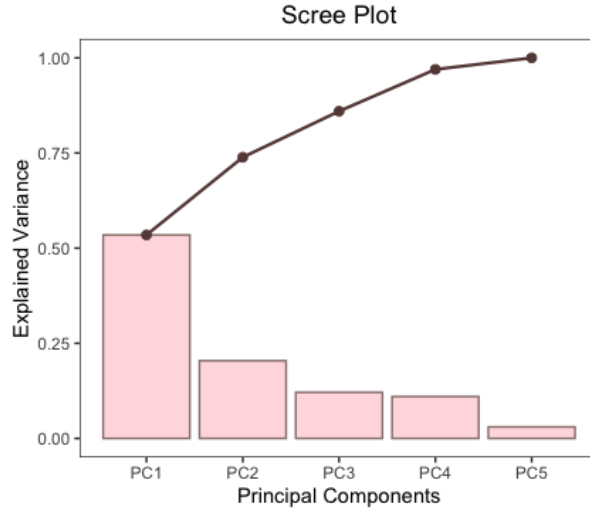


Figure 5: Scree plot of explained and cumulative variance.

The loadings highlight the role of each variable in shaping the components (Table 2 and Figure 6)[cite: 1]:

- **PC1** contrasts **DistanceMRT** (positive loading) with **ConvenienceStores**, **Latitude**, and **Longitude** (negative loadings)[cite: 1]. This axis represents accessibility and location[cite: 1].
- **PC2** is dominated by **HouseAge** (loading = 0.975), capturing property age independently from location[cite: 1].

Table 2: Loadings on the first two principal components

Variable	PC1 (53.5%)	PC2 (20.4%)
HouseAge	-0.005	0.975
DistanceMRT	0.570	0.065
ConvenienceStores	-0.461	0.102
Latitude	-0.450	0.125
Longitude	-0.509	-0.138

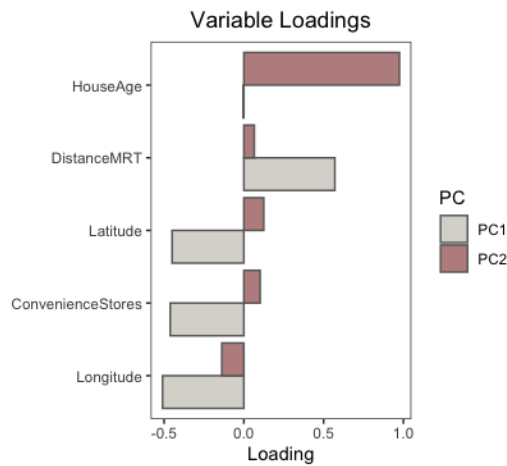
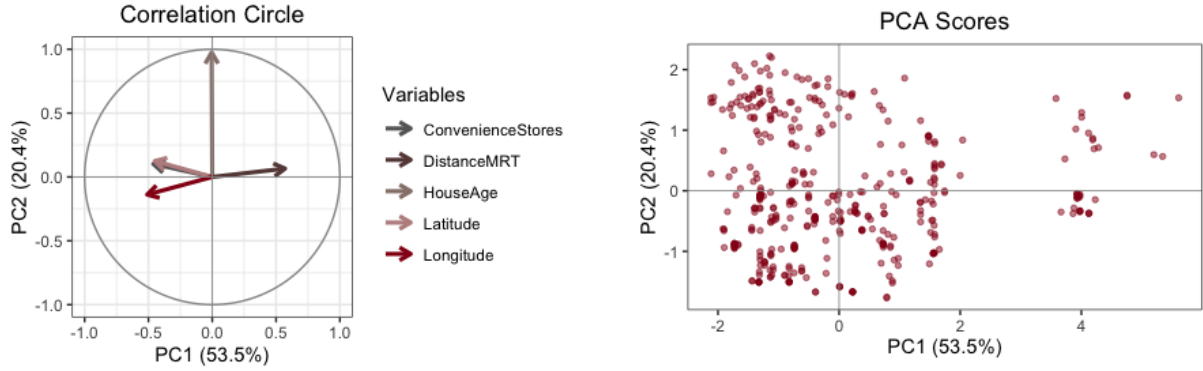


Figure 6: Variable Loadings on PC1 and PC2.



(a) Correlation Circle (PC1–PC2)

(b) Score Plot

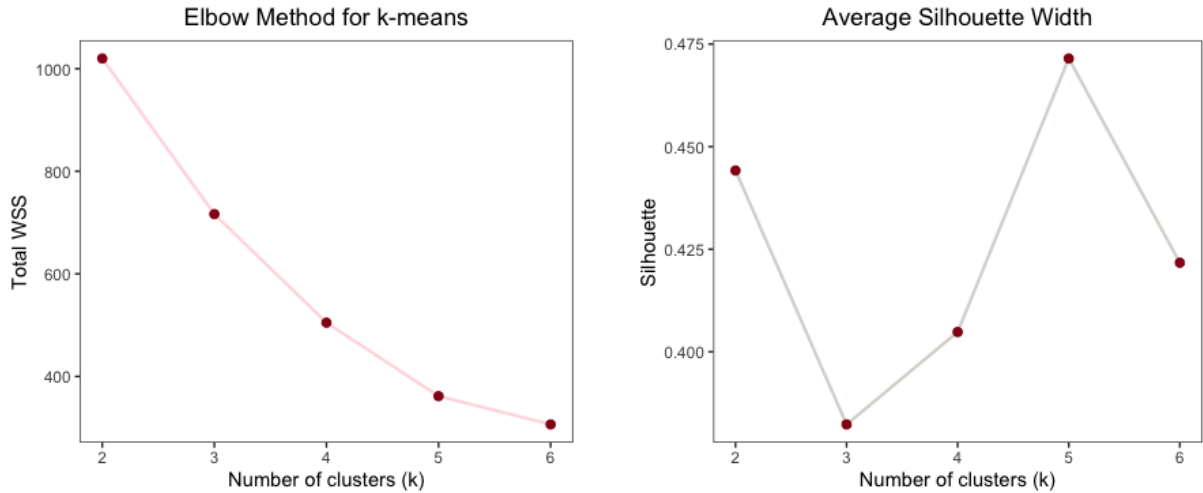
Figure 7: PCA representation in the first two principal dimensions.

The correlation circle (Figure 7a) illustrates these relationships: **HouseAge** aligns closely with PC2, while **DistanceMRT** extends in the positive PC1 direction, opposed to **Latitude**, **Longitude**, and **ConvenienceStores**[cite: 1]. The score plot (Figure 7b) shows property transactions distributed according to accessibility (PC1) and building age (PC2)[cite: 1].

6 Clustering Results

To explore potential segmentation in the housing market, *k*-means clustering was applied on the first three principal components of the dataset[cite: 1]. Two complementary approaches were used to determine the appropriate number of clusters: the Elbow Method and Silhouette Analysis[cite: 1].

- The Elbow plot of the within-cluster sum of squares (WSS) (Figure 8a) suggested an inflection around $k = 4$ – 5 [cite: 1].
- The average silhouette width (Figure 8b) reached its maximum at $k = 5$ (0.471)[cite: 1].



(a) Elbow plot (Total WSS)

(b) Average Silhouette Width

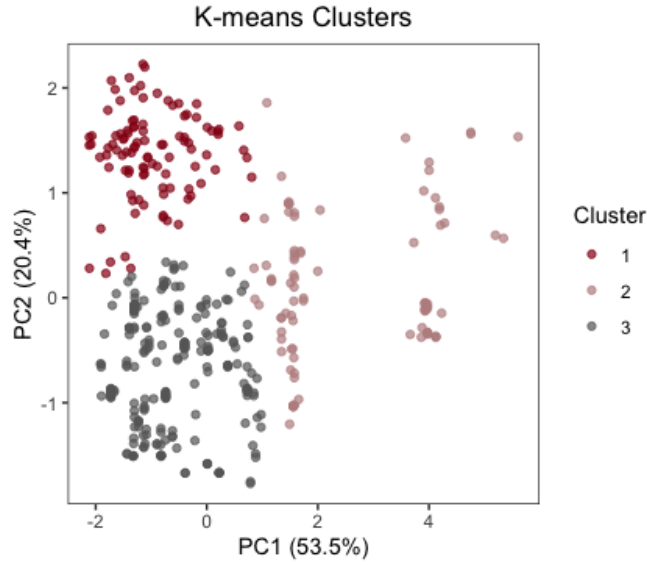
Figure 8: Cluster diagnostic plots for $k \in \{2, \dots, 6\}$.

However, in line with the principle of parsimony and economic interpretability, $k = 3$ was retained as the final solution[cite: 1].

Table 3: Cluster Profiles ($k = 3$)

Cluster	n	Mean Price	Mean Age	Mean Dist. MRT	Mean Stores
1	102	12.2	33.3	460	5.85
2	90	6.92	18.1	2991	1.52
3	222	13.0	10.4	597	4.33

- **Cluster 1:** older houses (≈ 33 years), close to MRT stations (≈ 460 m), with high density of convenience stores (≈ 6) and moderately high prices ($\approx 122,000$ NTD/m²)[cite: 1].
- **Cluster 2:** peripheral properties far from MRT stations (≈ 3 km), few nearby stores (≈ 1.5), medium age (≈ 18 years), and lowest prices ($\approx 69,200$ NTD/m²)[cite: 1].
- **Cluster 3:** the youngest houses (≈ 10 years), well connected (≈ 600 m), intermediate amenities (≈ 4.3), and highest average prices ($\approx 130,000$ NTD/m²)[cite: 1].

Figure 9: K -means clustering results ($k = 3$) projected onto PC1–PC2.

7 Data Preparation for Supervised Learning

Before fitting predictive models, the dataset was prepared for regression analysis[cite: 1]:

- Target: natural logarithm of the unit house price, $\text{LogPrice} = \log(\text{PricePerUnitArea})$ [cite: 1].
- Predictors: `HouseAge`, `DistanceMRT`, `ConvenienceStores`, `Latitude`, and `Longitude`[cite: 1].
- Train/test split: 70% train and 30% test with a fixed random seed (123)[cite: 1].
- Standardization ($\mu = 0, \sigma = 1$) was computed strictly on the training set and applied out-of-sample to avoid leakage[cite: 1].

8 Baseline Model: Linear Regression

An Ordinary Least Squares (OLS) model was estimated on the training set using standardized predictors[cite: 1].

Table 4: Performance of the Baseline Linear Regression Model

Dataset	R^2	RMSE
Train	0.713	0.198
Test	0.604	0.275

The model explains 71% of variance in the training set and 60% in the test set[cite: 1].

Table 5: Linear Regression Model Coefficients

Variable	Coefficient
(Intercept)	2.353
HouseAge	−0.063
DistanceMRT	−0.162
ConvenienceStores	0.076
Latitude	0.108
Longitude	0.023

- **House age** (−0.063) and **distance to MRT** (−0.162) show negative effects on prices[cite: 1].
- **Convenience stores** (+0.076) confirm the value of local amenities[cite: 1].
- **Geographical coordinates** reflect positive trends toward eastern and north-eastern districts of Taipei (such as Xinyi and Da'an)[cite: 1].

9 Neural Network (MLP)

An MLP with 1 hidden layer (5 neurons) and weight decay 0.01 was trained for 500 iterations[cite: 1].

Table 6: Performance of Neural Network (MLP)

Dataset	R^2	RMSE
Train	0.845	0.145
Test	0.659	0.256

The neural network improves out-of-sample accuracy compared to linear regression ($R^2 = 0.659$, RMSE = 0.256)[cite: 1].

10 Random Forest

A Random Forest regressor with 500 trees and $m_{\text{try}} = 2$ was trained on raw predictors[cite: 1].

Table 7: Random Forest Performance

Dataset	R^2	RMSE
Train	0.951	0.081
Test	0.703	0.238

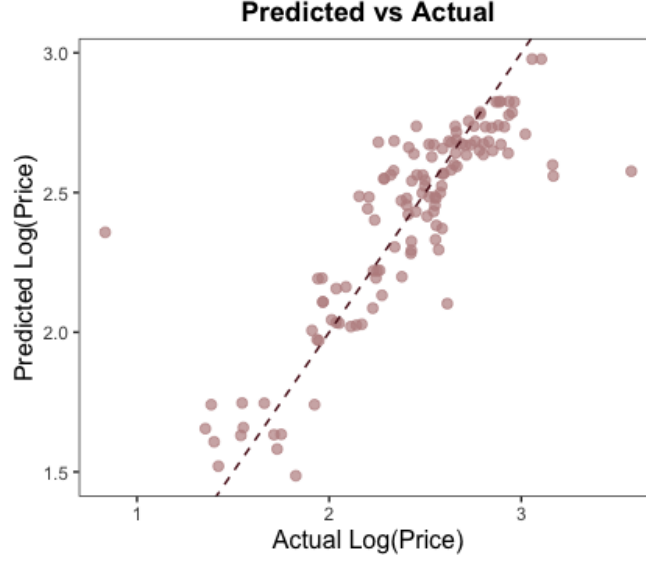


Figure 10: Random Forest: Predicted vs Actual log-price (Test set).

Distance to MRT is the most important feature, followed by coordinates, with house age and convenience stores contributing moderately[cite: 1].

11 Model Comparison

Table 8: Model Performance Comparison on Test Set

Model	R^2 Test	RMSE Test
Random Forest	0.703	0.238
Neural Network (MLP)	0.659	0.256
Linear Regression	0.604	0.275

1. **Random Forest is the best-performing model** with the lowest RMSE (0.238) and highest R^2 (0.703)[cite: 1].
2. **Neural Network performs reasonably well** (RMSE = 0.256), capturing non-linear patterns[cite: 1].
3. **Linear Regression is the baseline benchmark** (RMSE = 0.275)[cite: 1].

12 Conclusion

This study examined the housing market of Taipei using unsupervised and supervised learning methods[cite: 1]. PCA and clustering identified three distinct market segments driven by accessibility, building age, and local amenities[cite: 1]. In the supervised stage, Random Forest achieved superior predictive capability by effectively capturing non-linear spatial dependencies[cite: 1].